Experiment:6 Simulation-Based Investigation of Microwave Semiconductor Devices and Frequency-Conversion Characteristics of a Microwave Mixer

Aim: To simulate microwave semiconductor devices using MATLAB and analyze the frequency-conversion characteristics of a microwave mixer.

Apparatus:

Computer/Laptop
MATLAB software

Procedure:

- Open MATLAB and create a new script.
- Define the RF signal, local oscillator (LO) signal, and their frequencies.
- Generate the RF and LO signals using sine wave equations.
- Simulate the mixer by multiplying the RF and LO signals.
- Calculate the sum and difference frequencies.
- Plot the input signals, mixer output, and frequency spectrum.
- Run the program and observe the frequency conversion characteristics

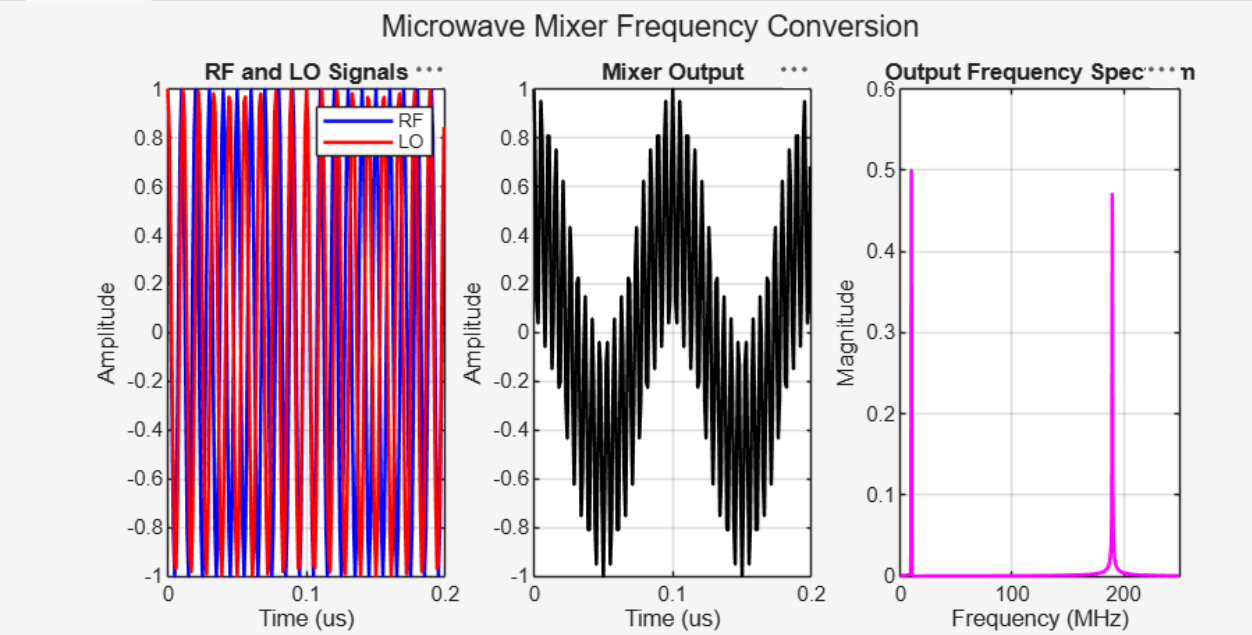
Code:

```
clc;
clear;
close all;
Fs = 1e9;
t = 0:1/Fs:2e-6;
fRF = 100e6;
fLO = 90e6;
```

```
RF = cos(2*pi*fRF*t);
LO = cos(2*pi*fLO*t);
Y = RF.*LO;
N = length(Y);
F = (0:N-1)*Fs/N;
S = abs(fft(Y))/N;
figure;
subplot(1,3,1);
plot(t(1:200)*1e6,RF(1:200),'b','LineWidth',1.5);
hold on;
plot(t(1:200)*1e6,LO(1:200),'r','LineWidth',1.5);
grid on;
xlabel('Time (us)');
ylabel('Amplitude');
title('RF and LO Signals');
legend('RF','LO');
subplot(1,3,2);
plot(t(1:200)*1e6,Y(1:200),'k','LineWidth',1.5);
grid on;
xlabel('Time (us)');
ylabel('Amplitude');
title('Mixer Output');
subplot(1,3,3);
plot(F(1:N/2)/1e6,2*S(1:N/2),'m','LineWidth',1.5);
grid on;
xlabel('Frequency (MHz)');
ylabel('Magnitude');
```

```
title('Output Frequency Spectrum');
xlim([0 250]);
sgtitle('Microwave Mixer Frequency Conversion');
fprintf('RF Frequency = %.0f MHz\n',fRF/1e6);
fprintf('LO Frequency = %.0f MHz\n',fLO/1e6);
fprintf('Difference Frequency = %.0f MHz\n',(fRF-fLO)/1e6);
fprintf('Sum Frequency = %.0f MHz\n',(fRF+fLO)/1e6);
```

Output:



Result:

The microwave mixer was simulated successfully using MATLAB. The output frequency spectrum shows the difference frequency of 10 MHz and the sum frequency of 190 MHz, demonstrating frequency conversion.